

Nuojunxi (Steven) Zhang

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Education

University of South Florida, Ph.D. in Computer Science & Engineering Aug 2026 – Present

- Bellini College of Artificial Intelligence, Cybersecurity and Computing
- Advisor: Prof. Yi Sheng

Kean University, B.S. in Computer Science Sept 2022 – May 2026

- Degree completed at Kean University, Union, NJ
- **Coursework:** Data Structures and Algorithms, Computer Architecture, Object-Oriented Programming
- Dean's Honor List, Kean University

Technologies

Languages: Java, Python, C#, SQL, MATLAB

Tools & Frameworks: PyTorch, TensorFlow, OpenCV, Hugging Face, Git, Linux, Docker, Unity, Object-Oriented Design, System Testing, Technical Writing

Publication

Segmenting What Matters: A Dual-Stage Active Learning Framework for Weakly Supervised Breast Ultrasound Segmentation October 2025

Nuojunxi Zhang, Meng Xu, Guanchao Tong, Kuan Huang
IEEE International Conference on Bioinformatics and Biomedicine (BIBM) 2025

Segmenting with Reliable Guidance: Dual-Branch AMR and CAM-Guided Learning for Weakly Supervised Breast Ultrasound Segmentation

Nuojunxi Zhang, Kuan Huang
Manuscript submitted to IEEE Journal of Biomedical and Health Informatics (JBHI)

Experience

Reviewer, *Pattern Recognition* (Elsevier) 2026

Graduate Research Assistant, YES Lab, University of South Florida Aug 2026 – Present

- Supervised by Prof. Yi Sheng, working on X-ray CT reconstruction and detector calibration
- First-author abstract under review at SPIE Medical Imaging 2027

Graduate Instructional Assistant, Bellini College of AI, Cybersecurity and Computing, University of South Florida Aug 2026 – Present

- Instructional assistant for a course on incident response methods
- Supporting labs and hands-on exercises, grading, and office hours

Research Assistant (NSF-Funded), AI4Healthcare Lab, Kean University June 2025 – Sept 2025

- Appointed under NSF funding, supervised by Prof. Kuan Huang
- Contributing to a journal-targeted research on ultrasound-based medical image segmentation
- Responsible for algorithm development, experimental design, and manuscript preparation

Research Assistant (NSF-Funded), AI4Healthcare Lab, Kean University Dec 2024 – May 2025

- Designed a dual-stage active learning framework for weakly supervised tumor segmentation using image-level ultrasound labels
- Developed HSV-based CAM filtering and SAM-guided pseudo-label refinement to enhance mask quality
- Integrated a Mean Teacher segmentation model with iterative uncertainty sampling, achieving 68.25% IoU and 79.39% DSC on the BUSI dataset
- Conducted end-to-end project design, experimentation, and publication as an NSF-funded research assistant

- **First-author paper accepted at IEEE BIBM 2025** (acceptance rate: 19.8%)

Research Participant, Medfusion Fake Image Generator, Kean University Sept 2024 – Dec 2024

- Developed a synthetic medical image pipeline using Variational Autoencoders (VAE) and Diffusion Models to augment breast ultrasound datasets for weakly supervised segmentation
- Training a latent VAE embedder and conditional diffusion model on BUSI dataset
- Designed experiments using diffusion-generated images and weak labels to evaluate their impact on downstream segmentation performance

Research Assistant, Vision-AI Lab, Wenzhou-Kean University Sept 2023 – May 2024

- Joined Prof. Gupta's Vision-AI Lab to support projects in computer vision and medical imaging
- Participated in a SPF-funded initiative focused on raw image denoising using Bayer pattern modeling
- Worked on latent space extensions and codebase adaptation based on a prior CVPR 2022 Workshop project developed by a former lab member under the same advisor

Vice President, Wenzhou-Kean University Computer Club Jan 2024 – June 2024

- Managed and planned club operations, overseeing departmental organization and project planning
- Led strategic initiatives that increased active membership and project participation by 30%

College Assistant, College of Science, Mathematics and Technology, Wenzhou-Kean University Aug 2022 – May 2024

- Assisted faculty with event promotion and day-to-day coordination of academic activities
- Streamlined scheduling workflows, improving operational efficiency across departmental projects

Research Assistant, Parkinson's Rehabilitation System, Wenzhou Kean University Feb 2023 – June 2023

- Utilizing OpenCV and Mediapipe, I developed a program that captures patients' hand movements and provides real-time feedback for rehabilitation assessment
- Earned the *Outstanding Presentation* award at WKU Research Day for demonstrating prototype results

Projects

DSAL: Dual-Stage Active Learning for Medical Segmentation *Segmentation, Medical Imaging*

- Proposed a dual-stage framework combining weakly supervised segmentation with active learning
- Achieved 68.25% IoU and 79.39% DSC on BUSI dataset; reduced annotation costs.
- Published as **First Author** paper at **IEEE BIBM 2025**.

arXivPush: Automated arXiv Paper Summarization and Daily Digest System *LLMs, NLP Pipeline*

- Designed an end-to-end system that automatically fetches, deduplicates, summarizes, and translates new arXiv papers into bilingual daily digests.
- Integrated Discord API, Python automation, and LLM summarization (Ollama + Qwen2.5) for seamless academic community delivery.
- Released as an open-source project on GitHub, earning **29+ stars**, and reaching over **6.8K views** and **1300+ engagements** across communities.
- Companion iOS client currently in **TestFlight** beta.

Medfusion-Fake-Image Generator (Extension Project) *Diffusion Models, VAE, Medical Imaging*

- Extended the open-source Medfusion framework by integrating pathology-aware training for realistic, tumor-inclusive ultrasound image synthesis.
- Implemented 512×512 high-resolution generation with distributed training and customizable configurations for medical AI research.